Running the server (PowerShell Terminal / T1):

```
python proxy.py
```

```
Proxy running on port 10000...
```

Making a “curl” GET Request (Command Prompt Terminal / T2):

```
curl -v --proxy http://localhost:10000 http://example.com
```

T1 proxy terminal parses and displays request:

```
------ REQUEST ------
Time: 2026-04-27 17:12:28.537708
Client: 127.0.0.1:13356
Method: GET
Target: example.com:80
Path: /
User-Agent: curl/8.13.0
---------------------
```

Cache checks and returns MISS:

```
[2026-04-27 17:12:28.547711] Cache MISS: example.com:80/
```

Since MISS occurred, retrieve response from target server and return to client:

```
[2026-04-27 17:12:28.684701] Request forwarded to example.com
[2026-04-27 17:12:28.909007] Response received from server example.com; current size = 832B
[2026-04-27 17:12:28.911953] Response received from server example.com; current size = 837B
[2026-04-27 17:12:28.913949] Response from example.com:80 sent to this client: 127.0.0.1:13356
```

Display basic response parameters:

```
---- RESPONSE ----
Status: 200 OK
Size: 837 bytes
Time: 2026-04-27 17:12:28.918988
------------------
```

Curl (T2) logs:

```
* Host localhost:10000 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:10000...
*   Trying 127.0.0.1:10000...
* Connected to localhost (127.0.0.1) port 10000
* using HTTP/1.x
> GET http://example.com/ HTTP/1.1
> Host: example.com
> User-Agent: curl/8.13.0
> Accept: */*
> Proxy-Connection: Keep-Alive
```

```
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 27 Apr 2026 14:12:28 GMT
< Content-Type: text/html
< Transfer-Encoding: chunked
< Connection: close
< Server: cloudflare
< Last-Modified: Sat, 25 Apr 2026 11:01:06 GMT
< Allow: GET, HEAD
< Accept-Ranges: bytes
< Age: 2046
```

Then display the actual response body:

```
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
```

Connection then shuts down (curl specific behavior):

```
* shutting down connection #0
```

Proxy (T1) terminal then displays cache statistics:

```
---- Performance Summary ----
Total Requests: 1
Total Hits: 0
Total Misses: 1

Average Miss Time: 0.4013 ms
Average Hit Time: 0.0000 ms

Speedup Factor: ~No hits recorded
-----------------------------
```

Calling the same request again after 60 seconds results in the expiration logs:

```
[2026-04-27 17:22:24.606930] Entry expired, removing: example.com:80/
```

Then it’s fetched again from the target server with the same behavior as a cache MISS.

In case of HIT, simply logs it and doesn’t fetch it from target server:

```
[2026-04-27 18:20:04.466252] Cache HIT: example.com:80/
```

Finally, display performance summary:

```
---- Performance Summary ----
Total Requests: 4
Total Hits: 1
Total Misses: 3

Average Miss Time: 0.3151 ms
Average Hit Time: 0.0150 ms

Speedup Factor: ~21.007094227661657x
-----------------------------
```

In case of blocked IP Address:

```
Blocked the client from requesting anything since their address is blacklisted: 127.0.0.1
```

In case of blocked Host name:

```
Blocked the request attempt to example.com since it's blacklisted.
```

In case of blocked Port:

```
Blocked the request attempt to port 80 since it's blacklisted.
```

Curl command for HTTPS CONNECT method call:

```
curl -v -x http://localhost:10000 https://example.com
```

Logs for a CONNECT request:

```
------ REQUEST ------
Time: 2026-04-27 18:52:34.742739
Client: 127.0.0.1:14620
Method: CONNECT
Target: example.com:443
User-Agent: curl/8.13.0
--------------------

[2026-04-27 18:52:34.896089] Connection established with example.com:443; tunnel is ready.

[2026-04-27 18:52:35.580762] Tunnel closed for: example.com:443
```

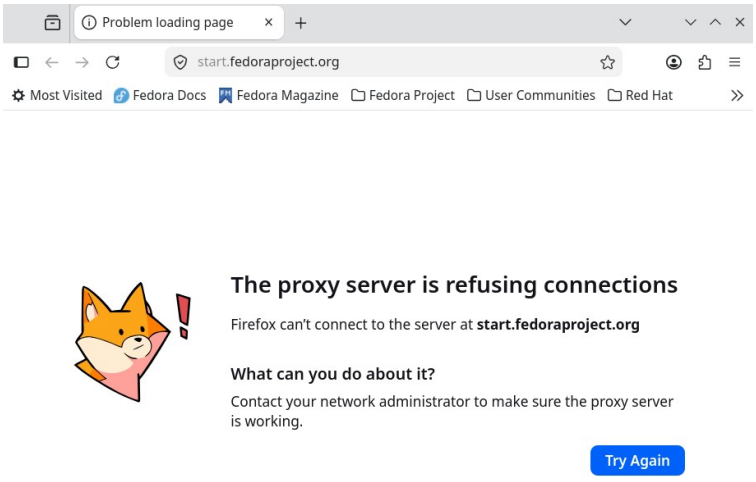
No response is logged since the proxy received it as encrypted and cannot access it, acts just as a tunnel for the data.

In case the cache was full, LRU algorithm is used, removes least used entry (first in ordered dictionary), and logs:

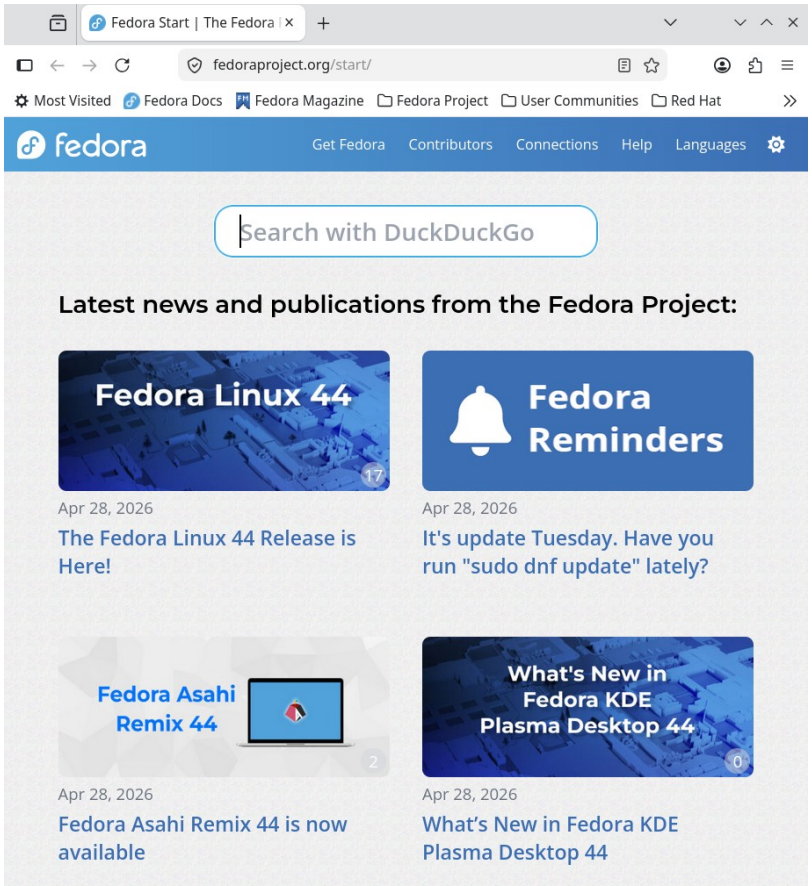
```
Cache full, least used entry removed from cache: example.com:80/1
```

In case of testing through browser (Firefox, with proxy configured to use the running server):

Server not yet running:



Website loaded after running the server:

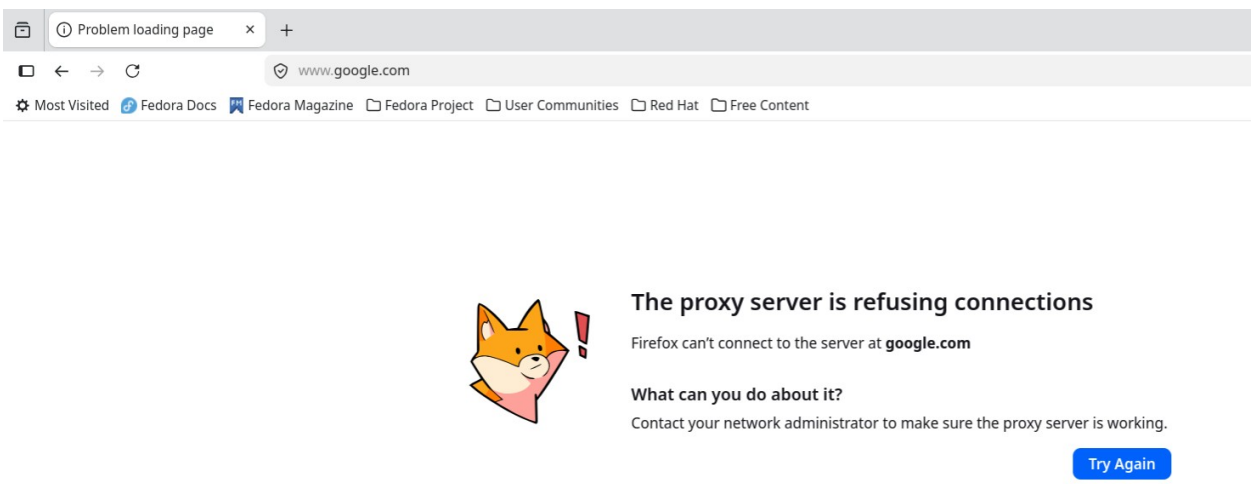


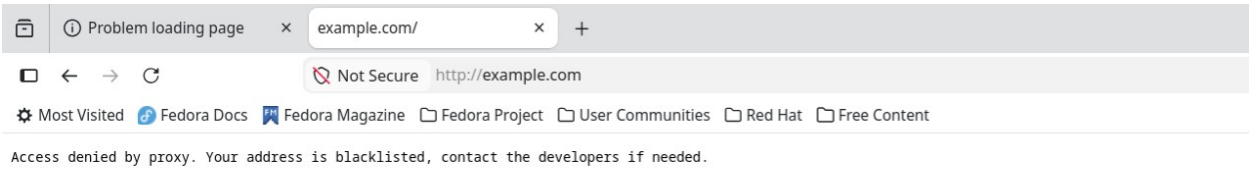
Proxy opens multiple tunnels to load the website:

```
------ REQUEST ------
Time: 2026-04-28 18:58:46.833393
Client: 127.0.0.1:56064
Method: CONNECT
Target: start.fedoraproject.org:443
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:149.0) Gecko/20100101 Firefox/149.0
--------------------

[2026-04-28 18:58:47.195203] Connection established with start.fedoraproject.org:443; tunnel is ready.

------ REQUEST ------
Time: 2026-04-28 18:58:47.756793
Client: 127.0.0.1:56070
Method: CONNECT
Target: fedoraproject.org:443
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:149.0) Gecko/20100101 Firefox/149.0
--------------------

[2026-04-28 18:58:47.935863] Connection established with fedoraproject.org:443; tunnel is ready.
```

```
[2026-04-28 18:58:57.581852] Connection established with fedoramagazine.org:443; tunnel is ready.
[2026-04-28 18:58:57.644370] Connection established with fedoramagazine.org:443; tunnel is ready.
[2026-04-28 18:58:57.647618] Connection established with fedoramagazine.org:443; tunnel is ready.
[2026-04-28 18:58:57.649651] Connection established with peertube.linuxrocks.online:443; tunnel is ready.
[2026-04-28 18:58:57.651470] Connection established with fedoramagazine.org:443; tunnel is ready.
[2026-04-28 18:58:57.653153] Connection established with fedoramagazine.org:443; tunnel is ready.
[2026-04-28 18:58:57.655051] Connection established with fedoramagazine.org:443; tunnel is ready.

[2026-04-28 18:58:58.622491] Tunnel closed for: fedoramagazine.org:443

[2026-04-28 18:58:58.624114] Tunnel closed for: fedoramagazine.org:443

[2026-04-28 18:58:58.625237] Tunnel closed for: fedoramagazine.org:443

[2026-04-28 18:58:58.629717] Tunnel closed for: fedoramagazine.org:443

[2026-04-28 18:58:58.921928] Tunnel closed for: fedoramagazine.org:443

[2026-04-28 18:58:59.341118] Tunnel closed for: avatars.discourse-cdn.com:443

[2026-04-28 18:59:02.967069] Tunnel closed for: start.fedoraproject.org:443

[2026-04-28 18:59:13.770734] Tunnel closed for: fedoraproject.org:443
```

Blocked host:




```
[2026-04-28 19:01:08.521435] Tunnel closed for: ac.duckduckgo.com:443
[2026-04-28 19:01:08.525902] Connection established with google.com:443; tunnel is ready.
[2026-04-28 19:01:08.768313] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:08.772371] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:08.777076] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:08.843742] Blocked the request attempt to www.google.com since it's blacklisted.
[2026-04-28 19:01:09.303471] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:09.305855] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:09.307441] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:11.915446] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:11.920124] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
[2026-04-28 19:01:11.929153] Blocked the request attempt to incoming.telemetry.mozilla.org since it's blacklisted.
```

Blocked IP address: